

Torricelli makes the first mercury barometer



Barometers

A barometer measures atmospheric pressure (the weight of air in the atmosphere). Until recently there were two kinds: mercury barometers (see p.79), invented in 1643 by Evangelista Torricelli (above), and aneroid barometers, invented in 1844. Today they are mostly electronic.

Naming clouds

We owe the names used today for different types of cloud to Luke Howard, a London pharmacist. He had a passion for meteorology and, in 1802, published a list of cloud types, giving them Latin names based on their characteristics.



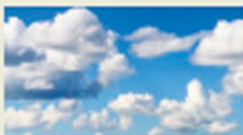
Cirrus (curl)

High, thin, wispy clouds composed of ice crystals that may indicate a change of weather.



Stratus (layer)

A low-level, flat blanket of cloud that produces overcast weather or light rain.



Cumulus (heap)

Puffy, cauliflower-like clouds that grow upward and may cause thunderstorms.



Scale registers contraction or expansion of hair

Hair hygrometer

Amazingly, a human hair can measure changes in air humidity (the amount of moisture in the air). In 1783, Swiss physicist Horace Bénédict de Saussure observed that hair lengthens on damp days and shrinks when it's dry. He put this fact to good use in making an instrument to measure humidity called a hair hygrometer.

Strand of hair

Hollow glass sphere



Cluster thermometer

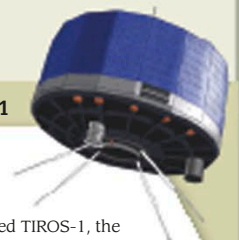
A thermometer is used to measure temperature. This 18th-century Italian thermometer is based on an invention of Galileo's. The six alcohol-filled tubes contain a number of small, hollow glass spheres. When the temperature rises, the alcohol expands, so its density decreases and the spheres sink.



Satellite image of Hurricane Andrew, 1992

Hurricane approach

Today, satellites collect and monitor weather data. Here, data from a NASA weather satellite was used to create this image of a hurricane moving to strike the coast of Florida. Meteorologists use satellites and computer modeling to give early warning of hurricanes.



TIROS-1

1849

The Smithsonian Institution established a weather observation network across the US. Hundreds of volunteers submitted reports by telegraph on a monthly basis.

1929

The first radiosonde—a box of instruments attached to an air balloon that collects meteorological information at high altitudes—was launched.

1953

The US National Hurricane Service began the system of assigning personal names to tropical storms originating in the Atlantic Ocean.

1960

NASA launched TIROS-1, the first successful weather satellite to provide accurate weather forecasts based on observations made from space.